

Adam Zhu

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EDUCATION

University of Texas at Austin

Austin, TX

Bachelor of Science in Computer Science, GPA 3.9/4.0

Expected Graduation - May 2027

Relevant Coursework: Artificial Intelligence, Operating Systems Honors, Computer Architecture Honors, Data Structures

EXPERIENCE

USA Computing Olympiad Mentor – Breakout Mentors | C++, Java, Python

Feb. 2025 – Present

- Improved algorithmic intuition of **Silver** students, increasing average coding and debugging speed by **20%** by introducing advanced algorithmic techniques including **binary search** and **graph traversal methods**.

Web Developer | JavaScript, React, Next.js, HTML, CSS, Tailwind, PostgreSQL, Express.js

Mar. 2023 – Jun. 2025

- Engineered **Arcane Odyssey Guides**, surpassing **3 million views** with an algorithm that sorts **40 billion** combinations in under **2 seconds** by streamlining frontend development with **React** and **Tailwind CSS** and optimizing the algorithm through fine tuning **heuristics** and strategic **pruning**.
- Designed and developed **Monkey Sense**, improving **500+** students' mental math speed and aptitude by **200%** by building a question generator, enhanced with a multiplayer duel system using an **Express.js** backend, a **PostgreSQL** leaderboard database, and a **WebSocket** client-server communication system.

Roblox and Unity Game Developer | Lua, C#, Roblox Studio, Unity, OOP

Mar. 2020 – Jul. 2025

- Published three Roblox titles (**Dig to India**, **Monster Survival**, **Cave Mining**) amassing **1.3M visits** and **10% 7-day retention** through mathematical game design and robust server security by encapsulating API calls.
- Developed **Geometry Battleground**, **CogQuest**, **The Past and The Curious**, and **Together In Spirit**, saving over **10,000** lines of code by leveraging Unity's object-oriented structure to create a **4-tier** class hierarchy.
- Achieved **#1 Overall** and **#1 Gameplay** in UT EGaDS! High Score Game Jam, **#1 People's Choice** and **#2 Best Narrative** in Texas Game Jam, and **#3** in BPA Software Engineering after presenting to industry engineers.

PROJECTS

AI Tetris Bot | Python, PyTorch, Reinforcement Learning, Vue.js, Flask

Apr. 2025 – Jul. 2025

- Designed and trained an **AI Tetris** bot using **Reinforcement Learning**, tweaking reward to survive **20x** longer.
- Built a frontend using **Vue.js** that instantly displays actions, using a **Neural Network** in a **Flask** backend.

Word Bomb AI Bot | Python, OpenCV, Tesseract, Computer Vision

Jan. 2025 – Feb. 2025

- Optimized real-time puzzle solving to under **2 seconds** by integrating **Computer Vision** with Tesseract OCR.
- Achieved **97%** text recognition accuracy by preprocessing image data using **NumPy** and **OpenCV**.

Clash Royale AI Deck Rater | Python, NumPy, Pandas, JavaScript, Keras

Apr 2024 – Dec 2024

- Improved model accuracy and reduced loss by **80%** through **machine learning optimization** in **Keras**.
- Curated a dataset of **2,000+** entries through **web scraping** and **data preprocessing** with JavaScript.

LeetCode Discord Bot | Python, GraphQL, PostgreSQL, APIs

Sep. 2024 – Oct. 2024

- Built a bot that seamlessly serves **100+** users, hosting an API using the **discord.py** and **Matplotlib** libraries.
- Reduced **GraphQL** API calls by **50%** using **PostgreSQL** database while keeping refreshes under **3 seconds**.

AWARDS

International Collegiate Programming Contest Regionals (sponsored by Jane Street) – 5th Place
Programming Titles: LeetCode Guardian (Top 0.37%), Codeforces Candidate Master (Top 2%)

UT Dallas Battle of the Brains (sponsored by State Farm) – 1st Place Advanced Division

UIL State – CS: 2x 1st Place Team, 3rd Individual; **Number Sense:** 4th Team, 8th Individual

TECHNICAL SKILLS

Languages: Java, C++, JavaScript/TypeScript, Python, C#, SQL, Lua, HTML/CSS, GraphQL

Frameworks: React, Next.js, Node, Express, Tailwind, NumPy, Pandas, OpenCV, .NET

Tools: Git, GitHub, Unity, Unreal, WSL, Agile/SCRUM, REST APIs, Postman

Topics: AI, Machine Learning, Computer Vision, Database, Full-Stack, OOP, Algorithms